

| No | Topic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Contact                          |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 02 | Internet of Things (IoT), Industry 4.0 and Smart System<br>- IoT for Smart Agriculture<br>- IoT for Smart Water Meter<br>- IoT for Egg Sorter<br>- IoT for Chicken Starter Cage<br>- IoT for Water Treatment Plan<br>- IoT for Fuel Management System<br>- IoT for Vessel Monitoring<br>- IoT for Fire Alarm and Evacuation System<br>- IoT for Water Density Monitoring System<br>- IoT for River Level Monitoring System<br>- IoT for Monitoring & Control of Temperature & Humidity Simultaneously in Modified HVAC System<br>- IoT for Energy Management System<br>- IoT for Room Control in Food Factory<br>- IoT for AI-Based Gas Leak Detector | Dr. Maman Budiman                |
| 03 | Development of Oil and Gas Pipeline Monitoring Using Fiber-Optic Distributed Acoustic Sensing and Deep-Learning-Based Signal Classification                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Prof. Bagus Endar B. Nurhandoko  |
| 04 | Digital Rock Physics<br>- Imaging: Processing and Analysis for Porous Media Analysis<br>- Machine Learning of Image Processing<br>- Development of in-situ XRM Uniaxial Comprehensive Test                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Dr. Fourier Dzar Eljabbar Latief |
| 05 | Rapid Identification of Vegetable Oils with Multi-Wavelength NIR LEDs Assisted by Machine Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Prof. Rachmat Hidayat            |
| 06 | Raman Quality Assurance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Prof. Mitra Djamal               |
| 07 | Machine Learning-Enhanced Selectivity of a Quartz Crystal Microbalance (QCM) Sensor Array                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Prof. Khairurrijal               |
| 08 | Biophysics and Medical Physics<br>- Physics-powered precision healthcare<br>- Monte Carlo simulation for precision dosimetry<br>- Medical imaging AI for clinical second opinion                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Dr. rer. nat. Freddy Haryanto    |
| 09 | From Waste to Value<br>- Thermal Optimization and Incineration Ash Valorization for Sustainable Resource Recovery                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Dr. Neni Surtiyeni               |

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|    | <ul style="list-style-type: none"> <li>- Thermophysical Engineering and Development of Incineration Ash-Based Materials</li> <li>- Environmental Performance &amp; Bottom Ash Characterization</li> </ul>                                                                                                                                                                    |                                |
| 10 | Chicken-Maggot Apartment: From Waste to Nutrition                                                                                                                                                                                                                                                                                                                            | Dr. rer. nat. Linus A. Pasasa  |
| 11 | Electronics and Instrumentation Physics Lab <ul style="list-style-type: none"> <li>- Advanced Electrospinning Systems and Nanofiber Technologies</li> <li>- Fabrication and Characterization of Functional Nanofibers</li> <li>- Fabrication of Nanofiber-Based Air Filter</li> </ul>                                                                                        | Dr. Irfan Dwi Aditya           |
| 12 | Many Entity Physical System <ul style="list-style-type: none"> <li>- ABM Simulation of Pile Stability and Angle of Repose of Granular Materials</li> <li>- Simulation of 2-D Hexagonal Close-Packed of Floating Sphere</li> <li>- ABM Simulation for Origin-Destination Matrix, Intercity Commuting Traveler, Rest Area Visiting Probability, and Hall Evacuation</li> </ul> | Dr. rer. nat. Sparisoma Viridi |

#### List of Laboratories

1. Instrumentation Physics Laboratory
2. Internet of Things and Sensor Laboratory
3. Advanced Computational Physics Laboratory
4. Earth Physics Laboratory
5. Photonics and Magnetism Laboratory
6. Material Physics and Energy Laboratory
7. Theoretical High Energy Physics Laboratory
8. Nuclear Physics Simulation Laboratory